

Отчёт по лабораторной работе 06

Простейший вариант

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Содержание

1	Цель работы	5
2	Задание	6
3	Теоретическое введение	7
4	Выполнение лабораторной работы	8
5	Выводы	37
	Список литературы	38

Список иллюстраций

4.1	Создание нового релиза v.6.0.0	8
4.2	Запуск Julia	9
4.3	Установка пакетов Julia	9
4.4	Установка пакетов	10
4.5	Установка пакетов	10
4.6	Установка пакетов	10
4.7	Запуск скрипта scripts/sirpetri_run.jl	11
4.8	Таблица скрипта scripts/sirpetri_run.jl	12
4.9	Таблица скрипта scripts/sirpetri_run.jl	13
4.10	График скрипта scripts/sirpetri_run.jl	14
4.11	График скрипта scripts/sirpetri_run.jl	14
4.12	jupyter-notebook от scripts/sirpetri_run.jl	15
4.13	Запуск скрипта scripts/sirpetri_scan_parameters.jl	15
4.14	Таблица скрипта scripts/sirpetri_scan_parameters.jl	16
4.15	jupyter-notebook от scripts/sirpetri_scan_parameters.jl	17
4.16	Запуск скрипта scripts/sirpetri_animate.jl	17
4.17	Таблица скрипта scripts/sirpetri_animate.jl	18
4.18	jupyter-notebook от scripts/sirpetri_animate.jl	19
4.19	Запуск скрипта scripts/sirpetri_report.jl	19
4.20	График скрипта scripts/sirpetri_report.jl	20
4.21	График скрипта scripts/sirpetri_report.jl	20
4.22	jupyter-notebook от scripts/sirpetri_report.jl	21
4.23	Установка пакетов	22
4.24	Создание производных форматов	23
4.25	Создание производных форматов	23
4.26	Создание производных форматов	24
4.27	Создание производных форматов	24
4.28	Создание производных форматов	25
4.29	Настройка _quarto.yml	26
4.30	Подключение файлов	27

Список таблиц

1 Цель работы

Реализовать эпидемиологическую модель SIR (Susceptible–Infectious–Recovered) с использованием сетей Петри, выполнить детерминированное и стохастическое моделирование, исследовать влияние параметров на динамику эпидемии.

2 Задание

Создать рабочий каталог и установить необходимые пакеты.

Выполнить предложенный код модели SIR.

Преобразовать код в литературный стиль (Literate Programming).

Сгенерировать из литературного кода: чистый код, Jupyter Notebook и документацию в формате Quarto.

Выполнить код из Jupyter Notebook.

Интегрировать документацию Quarto в отчёт.

Добавить в литературный код вычисления для нового набора параметров.

Повторить генерацию всех артефактов с новыми параметрами и выполнить их.

Интегрировать новую документацию Quarto в отчёт.

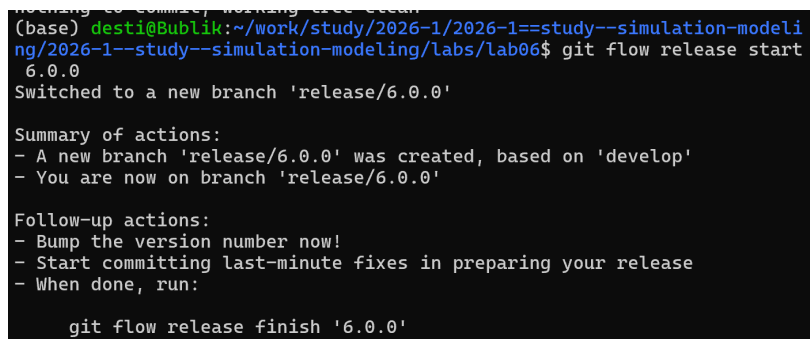
3 Теоретическое введение

Модель SIR описывает распространение эпидемии в популяции. Люди делятся на три группы: восприимчивые к болезни (S), инфицированные (I) и выздоровевшие с иммунитетом (R). Переходы между состояниями задаются двумя процессами: заражение ($S + I \rightarrow I + I$) с коэффициентом β и выздоровление ($I \rightarrow R$) с коэффициентом γ . Для моделирования используются сети Петри — математический аппарат для описания дискретных распределённых систем. В работе применяются два подхода: детерминированный (решение системы обыкновенных дифференциальных уравнений на основе закона действующих масс) и стохастический (прямой алгоритм Гиллеспи, учитывающий случайные флуктуации). Начальные условия: 990 восприимчивых, 10 заражённых, 0 выздоровевших. Исследуется влияние коэффициента заражения β на пик эпидемии и конечное число переболевших.

4 Выполнение лабораторной работы

4.0.1 1. Создание нового релиза

Создаем новый релиз v.6.0.0 и обновляем информацию по релизу v.4.0.0



```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06$ git flow release start
6.0.0
Switched to a new branch 'release/6.0.0'

Summary of actions:
- A new branch 'release/6.0.0' was created, based on 'develop'
- You are now on branch 'release/6.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

git flow release finish '6.0.0'
```

Рисунок 4.1: Создание нового релиза v.6.0.0

4.0.2 2. Настройка окружения Julia

Запускаем Julia и установку пакетов

4.0.3 3. Запуск скриптов и тема работы

Устанавливаем необходимые пакеты для первого скрипта

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. -e 'using Pkg; Pkg.add("AlgebraicPetri")'
Resolving package versions...
Installed LabelledArrays v1.19.0
Installed PrettyPrint v0.2.0
Installed GATlab v0.2.2
Installed RecursiveArrayTools v3.50.0
Installed AlgebraicPetri v0.10.0
Installed Permutations v0.4.23
Installed StructEquality v2.1.0
Installed NameResolution v0.1.5
Installed DiffEqBase v6.213.0
Installed StructTypes v1.11.0
Installed SciMLBase v2.153.1
Installed JSON3 v1.14.3
Installed MLStyle v0.4.17
Installed Catlab v0.17.5
Installed GeneralizedGenerated v0.3.3
Installed JuliaVariables v0.2.4
Installed XML2_jll v2.15.1+0
Installed AlgebraicInterfaces v0.1.4
Installed Comptime v0.1.2
Installed ChainRulesCore v1.26.1
Installed Compose v0.9.6
Installed LightXML v0.9.3
Installed PEG v1.0.4
[Warning: tarball content does not match git-tree-sha1 23/24
 @ Pkg.Operations /opt/julia-1.10.5/share/julia/stdlib/v1.10/Pkg/src/Operations.jl:646
Cloning [227ef7b5-1206-438b-ac65-934d6da304b8] ACSets from https://github.com/AlgebraicJulia/ACSets.jl.git
Installed ACSets v0.2.27
```

Рисунок 4.4: Установка пакетов

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. -e 'using Pkg; Pkg.add("Catlab")'
Resolving package versions...
Updating ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/Project.toml
[134e5e36] + Catlab v0.17.5
No Changes to ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/Manifest.toml
```

Рисунок 4.5: Установка пакетов

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. -e 'using Pkg; Pkg.add("OrdinaryDiffEq")'
Resolving package versions...
Updating ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/Project.toml
^ [1dea7af3] + OrdinaryDiffEq v6.105.0
No Changes to ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/Manifest.toml
```

Рисунок 4.6: Установка пакетов

Запуск первого скрипта scripts/sirpetri_run.jl

```
2026-1--study--simulation-modeling/labs/lab06/project/manifest.toml
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06/project$ julia --projec
t=. scripts/sirpetri_run.jl
Базовый прогон завершён. Результаты в data/ и plots/
```

Рисунок 4.7: Запуск скрипта scripts/sirpetri_run.jl

	A	B	C	D	E
1	time	S	I	R	
2	0	990	10	0	
3	0.5	2.47E-07	952.6914	47.30863	
4	1	3.28E-07	906.2281	93.77193	
5	1.5	-1.78E-07	862.0308	137.9692	
6	2	-2.66E-07	819.9891	180.0109	
7	2.5	2.39E-07	779.9977	220.0023	
8	3	-3.24E-07	741.9568	258.0432	
9	3.5	-2.31E-07	705.7711	294.2289	
10	4	3.04E-08	671.3503	328.6497	
11	4.5	3.63E-08	638.6081	361.3919	
12	5	-3.55E-07	607.4628	392.5372	
13	5.5	-3.49E-08	577.8365	422.1635	
14	6	2.30E-07	549.6551	450.3449	
15	6.5	3.25E-08	522.8481	477.1519	
16	7	-3.61E-07	497.3485	502.6515	
17	7.5	-2.04E-07	473.0925	526.9075	
18	8	4.39E-07	450.0195	549.9805	
19	8.5	1.58E-07	428.0718	571.9282	
20	9	-2.08E-07	407.1945	592.8055	
21	9.5	-2.45E-07	387.3354	612.6646	
22	10	1.22E-07	368.4448	631.5552	
23	10.5	1.74E-07	350.4756	649.5244	
24	11	6.84E-08	333.3827	666.6173	
25	11.5	-1.72E-07	317.1234	682.8766	
26	12	3.64E-07	301.6571	698.3429	
27	12.5	-3.93E-07	286.9451	713.0549	
28	13	-3.79E-07	272.9506	727.0494	
29	13.5	1.91E-07	259.6387	740.3613	
30	14	-2.26E-07	246.976	753.024	
31	14.5	4.24E-07	234.9308	765.0692	
32	15	-2.06E-08	223.4731	776.5269	
33	15.5	3.93E-07	212.5742	787.4258	
34	16	3.83E-07	202.2068	797.7932	
		sir_det			

Рисунок 4.8: Таблица скрипта scripts/sirpetri_run.jl

A1				time
	A	B	C	D
1	time	S	I	R
2	0	990	10	0
3	0.000219	989	11	0
4	0.000255	988	12	0
5	0.000435	987	13	0
6	0.001242	986	14	0
7	0.001373	985	15	0
8	0.001518	984	16	0
9	0.002194	983	17	0
10	0.002396	982	18	0
11	0.002458	981	19	0
12	0.002531	980	20	0
13	0.00275	979	21	0
14	0.00277	978	22	0
15	0.00292	977	23	0
16	0.00328	976	24	0
17	0.003389	975	25	0
18	0.003605	974	26	0
19	0.003676	973	27	0
20	0.003944	972	28	0
21	0.00406	971	29	0
22	0.004372	970	30	0
23	0.00448	969	31	0
24	0.004588	968	32	0
25	0.004731	967	33	0
26	0.004746	966	34	0
27	0.004977	965	35	0
28	0.004998	964	36	0
29	0.005012	963	37	0
30	0.005049	962	38	0
31	0.005081	961	39	0
32	0.005092	960	40	0
33	0.005164	959	41	0
34	0.005209	958	42	0
		sir_stoch		

Рисунок 4.9: Таблица скрипта scripts/sirpetri_run.jl

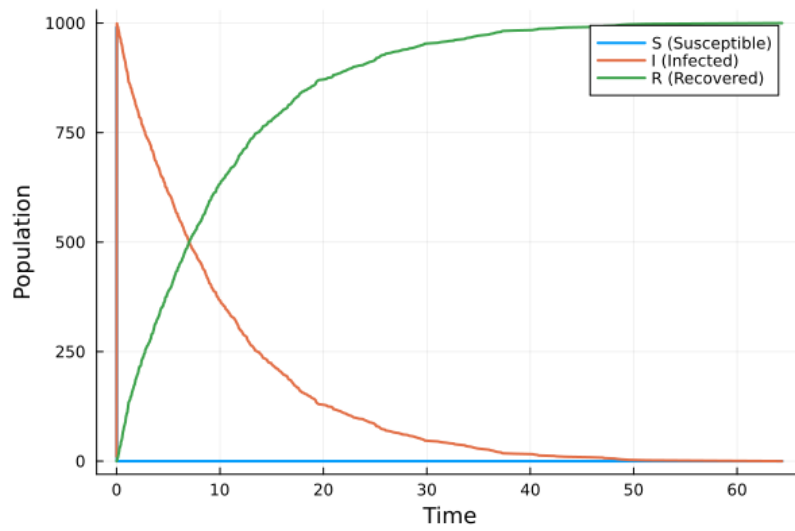


Рисунок 4.10: График скрипта scripts/sirpetri_run.jl

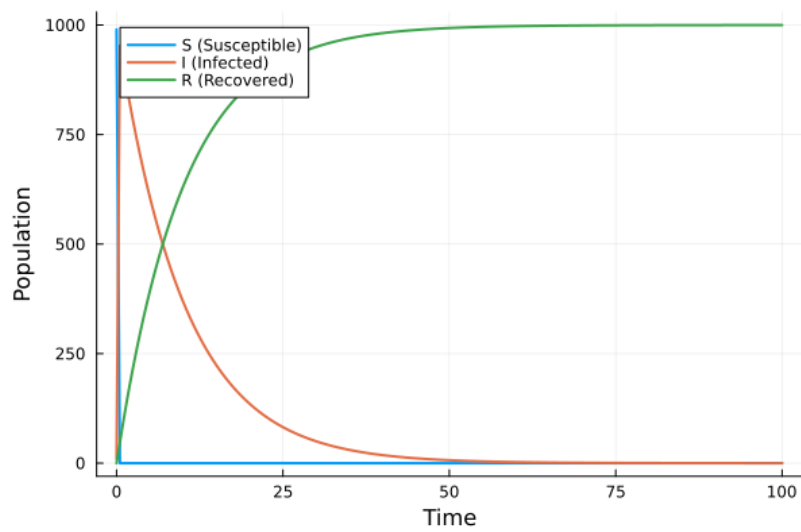


Рисунок 4.11: График скрипта scripts/sirpetri_run.jl

The screenshot shows a Jupyter Notebook interface with the title 'sirpetri_run'. The top bar includes 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', and 'Help' menus, along with a 'Not Trusted' warning and 'Julia 1.10' version indicator. The notebook contains three input cells and one output cell.

```

In [1]: using DrWatson
        @quickactivate "project"
        using Random
        include(scrdir("SIRPetri.jl"))
        using .SIRPetri
        using DataFrames, CSV, Plots

Параметры

In [2]: β = 0.3
        γ = 0.1
        tmax = 100.0

Out[2]: 100.0

Создаём сеть

In [3]: net, u0, states = build_sir_network(β, γ)

Out[3]: (AlgebraicPetri.LabelledPetriNet:
         T = 1:2
         S = 1:3
         I = 1:3
         O = 1:3
         Name = 1:0
         it : I → T = [1, 1, 2]
         is : I → S = [1, 2, 2]
         ot : O → T = [1, 1, 2]
         os : O → S = [2, 2, 3]
         tname : T → Name = [:infection, :recovery]
         sname : S → Name = [:S, :I, :R], [990.0, 10.0, 0.0], [:S, :I, :R])

Детерминированная симуляция

In [4]: df_det = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, n

```

Рисунок 4.12: jupyter-notebook от scripts/sirpetri_run.jl

Создание, запуск следующего скрипта scripts/sirpetri_scan_parameters.jl и создание его производных форматов

The terminal shows a user running a Julia script. The command is: `cd ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project; julia --project=. scripts/sirpetri_scan_parameters.jl`. The output indicates that the scanning process is complete and the result is saved in `data/sir_scan.csv`.

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_scan_parameters.jl
Сканирование β завершено. Результат в data/sir_scan.csv

```

Рисунок 4.13: Запуск скрипта scripts/sirpetri_scan_parameters.jl

	A	B	C	D
1	OI	peak_I	final_R	
2	0.1	955.626	999.9544	
3	0.15	954.1569	999.9545	
4	0.2	953.4237	999.9545	
5	0.25	952.9842	999.9545	
6	0.3	952.6914	999.9545	
7	0.35	952.4823	999.9545	
8	0.4	952.3256	999.9545	
9	0.45	952.2037	999.9545	
10	0.5	952.1062	999.9546	
11	0.55	952.0264	999.9546	
12	0.6	951.96	999.9546	
13	0.65	951.9037	999.9546	
14	0.7	951.8555	999.9546	
15	0.75	951.8138	999.9546	
16	0.8	951.7772	999.9546	
17				

Рисунок 4.14: Таблица скрипта scripts/sirpetri_scan_parameters.jl

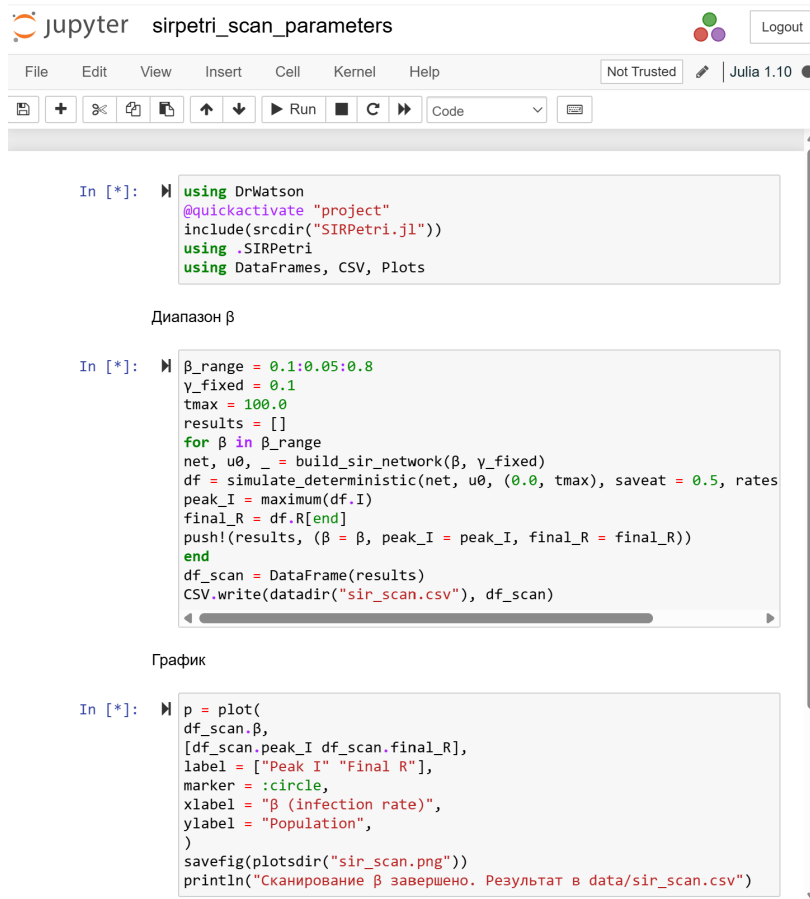


Рисунок 4.15: jupyter-notebook от scripts/sirpetri_scan_parameters.jl

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт scripts/sirpetri_animate.jl

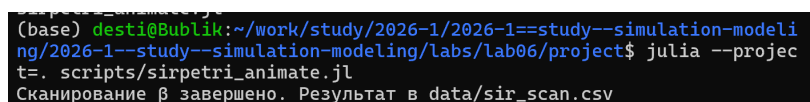


Рисунок 4.16: Запуск скрипта scripts/sirpetri_animate.jl

	A	B	C	
	Ol	peak_l	final_R	
	0.1	955.626	999.9544	
	0.15	954.1569	999.9545	
	0.2	953.4237	999.9545	
	0.25	952.9842	999.9545	
	0.3	952.6914	999.9545	
	0.35	952.4823	999.9545	
	0.4	952.3256	999.9545	
	0.45	952.2037	999.9545	
0	0.5	952.1062	999.9546	
1	0.55	952.0264	999.9546	
2	0.6	951.96	999.9546	
3	0.65	951.9037	999.9546	
4	0.7	951.8555	999.9546	
5	0.75	951.8138	999.9546	
6	0.8	951.7772	999.9546	
7				

Рисунок 4.17: Таблица скрипта scripts/sirpetri_animate.jl

The screenshot shows a Jupyter Notebook interface with the title 'sirpetri_animate'. The top bar includes 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', and 'Help' menus, along with a 'Not Trusted' status and 'Julia 1.10' version. The notebook contains three input cells:

```

In [1]: using DrWatson
        @quickactivate "project"
        include(sourcedir("SIRPetri.jl"))
        using .SIRPetri
        using DataFrames, CSV, Plots

        Диапазон  $\beta$ 

In [2]:  $\beta\_range = 0.1:0.05:0.8$ 
         $y\_fixed = 0.1$ 
         $tmax = 100.0$ 
         $results = []$ 
        for  $\beta$  in  $\beta\_range$ 
            net,  $u0, _ = build\_sir\_network(\beta, y\_fixed)$ 
             $df = simulate\_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates$ 
             $peak\_I = maximum(df.I)$ 
             $final\_R = df.R[end]$ 
            push!(results, ( $\beta = \beta$ ,  $peak\_I = peak\_I$ ,  $final\_R = final\_R$ ))
        end
         $df\_scan = DataFrame(results)$ 
        CSV.write(datadir("sir_scan.csv"), df_scan)

Out[2]: "/home/desti/work/study/2026-1/2026-1==study--simulation-modeling/20
26-1--study--simulation-modeling/labs/lab06/project/data/sir_scan.cs
v"

        График

In [*]:  $p = plot($ 
         $df\_scan.\beta,$ 
         $[df\_scan.peak\_I df\_scan.final\_R],$ 
         $label = ["Peak I" "Final R"],$ 
         $marker = :circle,$ 
         $xlabel = "\beta (infection rate)",$ 
         $ylabel = "Population"$ 

```

Рисунок 4.18: jupyter-notebook от scripts/sirpetri_animate.jl

Далее снова перехожу в директорию проекта и создаю и запускаю следующий скрипт scripts/sirpetri_report.jl

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06/project$ julia --projec
t=. scripts/sirpetri_report.jl
Отчётные графики сохранены в plots/

```

Рисунок 4.19: Запуск скрипта scripts/sirpetri_report.jl

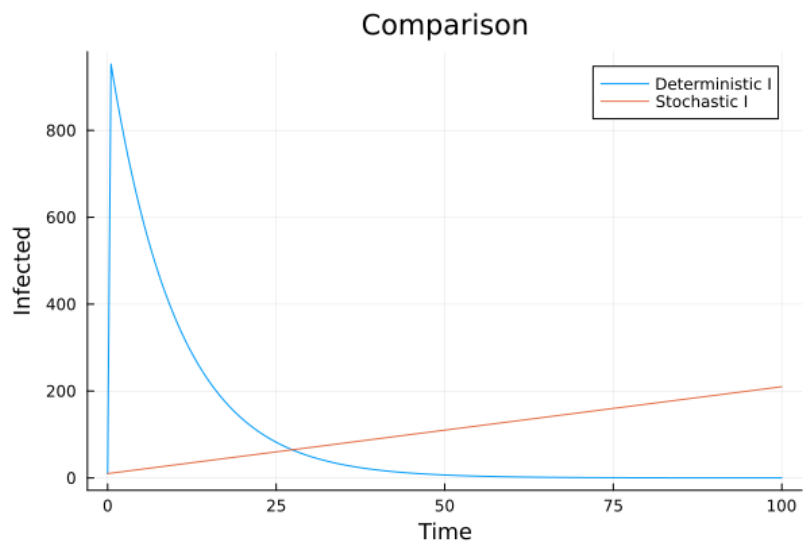


Рисунок 4.20: График скрипта `scripts/sirpetri_report.jl`

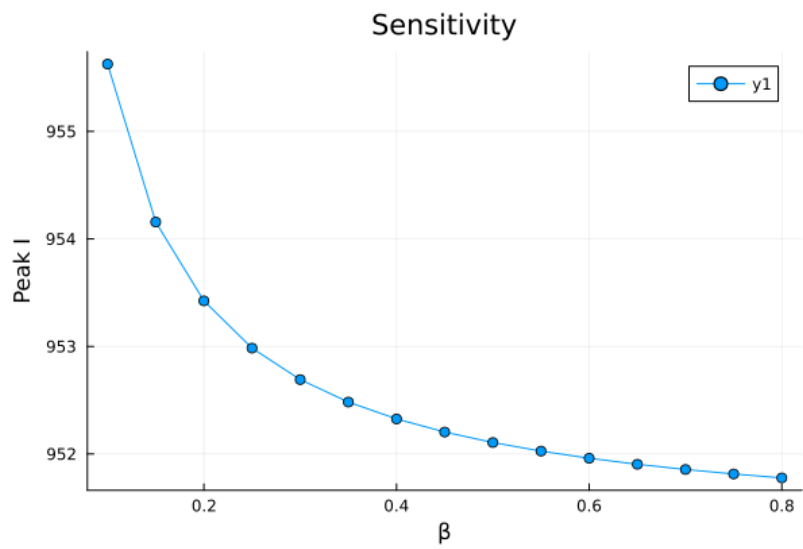


Рисунок 4.21: График скрипта `scripts/sirpetri_report.jl`

The screenshot shows a Jupyter Notebook titled "sirpetri_report" with unsaved changes. The interface includes a menu bar (File, Edit, View, Insert, Cell, Kernel, Help), a toolbar with icons for file operations and execution, and a status bar indicating "Not Trusted" and "Julia 1.10".

The first code cell (In [1]) contains the following Julia code:

```

using DrWatson
@quickactivate "project"
using DataFrames, CSV, Plots
df_det = CSV.read(datadir("sir_det.csv"), DataFrame)
df_stoch = CSV.read(datadir("sir_stoch.csv"), DataFrame)
df_scan = CSV.read(datadir("sir_scan.csv"), DataFrame)

```

The output (Out[1]) is a 15x3 DataFrame:

Row	β	peak_I	final_R
	Float64	Float64	Float64
1	0.1	955.626	999.954
2	0.15	954.157	999.954
3	0.2	953.424	999.954
4	0.25	952.984	999.955
5	0.3	952.691	999.955
6	0.35	952.482	999.955
7	0.4	952.326	999.955
8	0.45	952.204	999.955
9	0.5	952.106	999.955
10	0.55	952.026	999.955
11	0.6	951.96	999.955
12	0.65	951.904	999.955
13	0.7	951.856	999.955
14	0.75	951.814	999.955
15	0.8	951.777	999.955

Below the table, the text "Сравнение детерминированной и стохастической динамики" is displayed.

The second code cell (In [2]) contains the following Julia code:

```

p1 = plot(
df_det.time,
[df_det.I df_stoch.I[1:length(df_det.time)]],
label = ["Deterministic I" "Stochastic I"],
xlabel = "Time",
ylabel = "Infected",
title = "Comparison",
)
savefig(plotsdir("comparison.png"))

```

The output (Out[2]) is the file path: `"/home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--s"`.

Рисунок 4.22: jupyter-notebook от scripts/sirpetri_report.jl

Устанавливаем необходимые пакеты для производных форматов

```

In [1]: using DrWatson
        @quickactivate "project"
        include(sourcedir("SIRPetri.jl"))
        using .SIRPetri
        using DataFrames, CSV, Plots

Диапазон  $\beta$ 

In [2]: beta_range = 0.1:0.05:0.8
        y_fixed = 0.1
        tmax = 100.0
        results = []
        for beta in beta_range
            net, u0, _ = build_sir_network(beta, y_fixed)
            df = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates
            peak_I = maximum(df.I)
            final_R = df.R[end]
            push!(results, (beta = beta, peak_I = peak_I, final_R = final_R))
        end
        df_scan = DataFrame(results)
        CSV.write(datadir("sir_scan.csv"), df_scan)

Out[2]: "/home/desti/work/study/2026-1/2026-1==study--simulation-modeling/20
        26-1--study--simulation-modeling/labs/lab06/project/data/sir_scan.cs
        v"

График

In [*]: p = plot(
        df_scan.beta,
        [df_scan.peak_I df_scan.final_R],
        label = ["Peak I" "Final R"],
        marker = :circle,
        xlabel = "beta (infection rate)",
        ylabel = "Peak I"
    )
  
```

Рисунок 4.23: Установка пакетов

Создаем производные форматы от скриптов

```

@ Base ./client.jl:552
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl src/SIRPetri.jl
Генерация из: src/SIRPetri.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/src/SIRPetri.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/SIRPetri/SIRPetri.jl`
      √ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/SIRPetri/SIRPetri.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/src/SIRPetri.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/SIRPetri/SIRPetri.qmd`
      √ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/SIRPetri/SIRPetri.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/src/SIRPetri.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/SIRPetri/SIRPetri.ipynb`
      √ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/SIRPetri/SIRPetri.ipynb
Готово! Все файлы созданы.

```

Рисунок 4.24: Создание производных форматов

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_run.jl
Генерация из: scripts/sirpetri_run.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run/sirpetri_run.jl`
      √ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run/sirpetri_run.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_run/sirpetri_run.qmd`
      √ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_run/sirpetri_run.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_run/sirpetri_run.ipynb`
      √ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_run/sirpetri_run.ipynb
Готово! Все файлы созданы.

```

Рисунок 4.25: Создание производных форматов

```

Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06/project$ julia --projec
t. scripts/tangle.jl scripts/sirpetri_scan_parameters.jl
Генерация из: scripts/sirpetri_scan_parameters.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/
project/scripts/sirpetri_scan_parameters.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/
sirpetri_scan_parameters/sirpetri_scan_parameters.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulatio
n-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts
/sirpetri_scan_parameters/sirpetri_scan_parameters.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/proj
ect/scripts/sirpetri_scan_parameters.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown
/sirpetri_scan_parameters/sirpetri_scan_parameters.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-model
ing/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpe
tri_scan_parameters/sirpetri_scan_parameters.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/s
cripts/sirpetri_scan_parameters.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebook
s/sirpetri_scan_parameters/sirpetri_scan_parameters.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-mod
eling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/si
rpetri_scan_parameters/sirpetri_scan_parameters.ipynb

Готово! Все файлы созданы.

```

Рисунок 4.26: Создание производных форматов

```

Готово! Все файлы созданы.
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06/project$ julia --projec
t. scripts/tangle.jl scripts/sirpetri_animate.jl
Генерация из: scripts/sirpetri_animate.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/
project/scripts/sirpetri_animate.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/
sirpetri_animate/sirpetri_animate.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulatio
n-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts
/sirpetri_animate/sirpetri_animate.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/proj
ect/scripts/sirpetri_animate.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown
/sirpetri_animate/sirpetri_animate.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-model
ing/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpe
tri_animate/sirpetri_animate.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/s
cripts/sirpetri_animate.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebook
s/sirpetri_animate/sirpetri_animate.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-mod
eling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/si
rpetri_animate/sirpetri_animate.ipynb

Готово! Все файлы созданы.

```

Рисунок 4.27: Создание производных форматов


```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_report.jl
Генерация из: scripts/sirpetri_report.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report/sirpetri_report.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report/sirpetri_report.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_report/sirpetri_report.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_report/sirpetri_report.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_report/sirpetri_report.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_report/sirpetri_report.ipynb
Готово! Все файлы созданы.

```

Рисунок 4.28: Создание производных форматов

В каталоге отчёта в файл `_quarto.yml` включаем поддержку кода `julia`

```

!_quarto.yml X
Ubuntu > home > desti > work > study > 2026-1 > 2026-1==study--simulation-1

1 project:
2   title: "report"
3   output-dir: "_output"
4
5 standalone: true
6 self-contained: true
7
8 ## Julia support
9 # engine: julia
10 # julia:
11 #   execlags: ["--project=../project"]
12
13 ## Generic options
14 lang: ru-RU
15 number-sections: true
16 toc: true
17 toc-title: "Содержание"
18 toc-depth: 2
19 ## Crossref customization
20 crossref:
21   lof-title: "Список иллюстраций"
22   lot-title: "Список таблиц"
23   lol-title: "Листинги"
24 ## Bibliography
25 bibliography:
26   bib: bib/cite.bib
27   csl: _resources/csl/gost-r-7-0-5-2008-numeric.csl
28 ## Formats
29 format:
30   ### Docx output format
31   docx:
32     toc: true
33     number-sections: true
34     toc-depth: 2
35   ### Pdf output format
36   pdf:
37     # pdf-engine: xelatex
38     ### minted support
39     # filters:
40     #   - minted-quarto
41     toc: true
42     number-sections: true
43     colorlinks: false
44     toc-depth: 2
45     lof: true # List of figures
46     lot: true # List of tables
47     #### Document
48     documentclass: scrreprt
49     papersize: a4
50     fontsize: 12pt
51     linestretch: 1.5
52     #### Language
53     babel-lang: russian
54     babel-otherlangs: english
55     #### Biblatex
56     cite-method: biblatex
57     biblio-style: gost-numeric
58     biblatexoptions:
59       - backend=biber
60       - langhook=extras
61       - autolang=other*
62     #### Misc options
63     csquotes: true
64     indent: true
65     include-in-header:
66       - file: _resources/tex/preamble.tex
67

```

Рисунок 4.29: Настройка _quarto.yml

В файле отчёта подключаем файл описания программ

```
{{< include ../project/markdown/SIRPetri/SIRPetri.qmd >}}
{{< include ../project/markdown/sirpetri_animate/sirpetri_animate.qmd >}}
{{< include ../project/markdown/sirpetri_report/sirpetri_report.qmd >}}
{{< include ../project/markdown/sirpetri_run/sirpetri_run.qmd >}}
{{< include ../project/markdown/sirpetri_scan_parameters/sirpetri_scan_paramet
ers.qmd >}}
```

Рисунок 4.30: Подключение файлов

```
module SIRPetri
using AlgebraicPetri
using Catlab.CategoricalAlgebra
using Catlab.Graphics
using OrdinaryDiffEq
using Plots
using DataFrames
using Random

export build_sir_network, simulate_deterministic, simulate_stochastic
export plot_sir, to_graphviz_sir
"""
build_sir_network(β=0.3, γ=0.1)
Создаёт размеченную сеть Петри для модели SIR.
Возвращает (net::LabelledPetriNet, u0::Vector{Float64}, states::Vector{Symbol})
"""
function build_sir_network(β = 0.3, γ = 0.1)
states = [:S, :I, :R]
```

Создаём сеть Петри, описывая переходы напрямую Переход infection: $S + I \rightarrow I + I$ Переход recovery: $I \rightarrow R$

```
net = LabelledPetriNet(
states,
:infection => (:S, :I) => (:I, :I)),
:recovery => (:I) => (:R)),
)
```

Начальная маркировка: S=990, I=10, R=0

```
u0 = [990.0, 10.0, 0.0]
return net, u0, states
end
"""
sir_ode(net, rates)
Возвращает функцию правой части ОДУ для сети Петри.
Используется для детерминированной симуляции.
"""
function sir_ode(net, rates = [0.3, 0.1])
function f!(du, u, p, t)
S, I, R = u
β, γ = rates
```

Пропускные способности (закон действующих масс)

```
infection_rate = β * S * I
recovery_rate = γ * I
```

Изменения

```
du[1] = -infection_rate # dS/dt
du[2] = infection_rate - recovery_rate # dI/dt
```

```

du[3] = recovery_rate # dR/dt
end
return f!
end
"""

simulate_deterministic(net, u0, tspan; saveat=0.1, rates=[0.3,0.1])
Выполняет детерминированную ODE-симуляцию.
Возвращает DataFrame с колонками time, S, I, R.
"""

function simulate_deterministic(net, u0, tspan; saveat = 0.1, rates = [0.3, 0.1])
f = sir_ode(net, rates)
prob = ODEProblem(f, u0, tspan)
sol = solve(prob, Tsit5(), saveat = saveat)
df = DataFrame(time = sol.t)
df.S = sol[1, :]
df.I = sol[2, :]
df.R = sol[3, :]
return df
end
"""

simulate_stochastic(net, u0, tspan; rates=[0.3,0.1], rng=Random.GLOBAL_RNG)
Стохастическая симуляция (алгоритм Гиллеспи) прямым методом.
Возвращает DataFrame.
"""

function simulate_stochastic(net, u0, tspan; rates = [0.3, 0.1], rng = Random.GLOBAL_
u = copy(u0)
t = 0.0
times = [t]

```

```

states = [copy(u)]
λ, μ = rates
while t < tspan[2]
  S, I, R = u
  a_inf = λ * S * I
  a_rec = μ * I
  a0 = a_inf + a_rec
  if a0 == 0
    break
  end
  dt = -log(rand(rng)) / a0
  r = rand(rng) * a0
  if r < a_inf

```

Заражение

```

u[1] -= 1
u[2] += 1
else

```

Выздоровление

```

u[2] -= 1
u[3] += 1
end
t += dt
if t <= tspan[2]
  push!(times, t)
  push!(states, copy(u))
end

```

```

end
df = DataFrame(time = times)
df.S = [s[1] for s in states]
df.I = [s[2] for s in states]
df.R = [s[3] for s in states]
return df
end
"""

plot_sir(df)
Строит график динамики S, I, R из DataFrame.
"""

function plot_sir(df)
p = plot(
df.time,
[df.S, df.I, df.R],
label = ["S (Susceptible)" "I (Infected)" "R (Recovered)"],
xlabel = "Time",
ylabel = "Population",
linewidth = 2,
)
return p
end
"""

to_graphviz_sir(net)
Визуализация сети Петри с помощью Graphviz.
"""

function to_graphviz_sir(net)
return to_graphviz(net, prog = "dot")

```

```
end
end # module
```

```
using DrWatson
@quickactivate "project"
include(sourcedir("SIRPetri.jl"))
using .SIRPetri
using DataFrames, CSV, Plots
```

Диапазон β

```
β_range = 0.1:0.05:0.8
β_fixed = 0.1
tmax = 100.0
results = []
for β in β_range
    net, u0, _ = build_sir_network(β, β_fixed)
    df = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [β, β_fixed])
    peak_I = maximum(df.I)
    final_R = df.R[end]
    push!(results, (β = β, peak_I = peak_I, final_R = final_R))
end
df_scan = DataFrame(results)
CSV.write(datadir("sir_scan.csv"), df_scan)
```

График

```
p = plot(
    df_scan.β,
    [df_scan.peak_I df_scan.final_R],
```



```

label = ["Peak I" "Final R"],
marker = :circle,
xlabel = "I (infection rate)",
ylabel = "Population",
)
savefig(plotsdir("sir_scan.png"))
println("Сканирование I завершено. Результат в data/sir_scan.csv")

```

```

using DrWatson
@quickactivate "project"
using DataFrames, CSV, Plots
df_det = CSV.read(datadir("sir_det.csv"), DataFrame)
df_stoch = CSV.read(datadir("sir_stoch.csv"), DataFrame)
df_scan = CSV.read(datadir("sir_scan.csv"), DataFrame)

```

Сравнение детерминированной и стохастической динамики

```

p1 = plot(
df_det.time,
[df_det.I df_stoch.I[1:length(df_det.time)]],
label = ["Deterministic I" "Stochastic I"],
xlabel = "Time",
ylabel = "Infected",
title = "Comparison",
)
savefig(plotsdir("comparison.png"))

```

Зависимость пика I от β

```

p2 = plot(
df_scan.ϕ,
df_scan.peak_I,
marker = :circle,
xlabel = "ϕ",
ylabel = "Peak I",
title = "Sensitivity",
)
savefig(plotsdir("sensitivity.png"))
println("Отчётные графики сохранены в plots/")

```

```

using DrWatson
@quickactivate "project"
using Random
include(sourcedir("SIRPetri.jl"))
using .SIRPetri
using DataFrames, CSV, Plots

```

Параметры

```

ϕ = 0.3
ϕ = 0.1
tmax = 100.0

```

Создаём сеть

```

net, u0, states = build_sir_network(ϕ, ϕ)

```

Детерминированная симуляция

```
df_det = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [β, γ])
CSV.write(datadir("sir_det.csv"), df_det)
```

Стохастическая симуляция

```
Random.seed!(123)
df_stoch = simulate_stochastic(net, u0, (0.0, tmax), rates = [β, γ])
CSV.write(datadir("sir_stoch.csv"), df_stoch)
```

Графики

```
p_det = plot_sir(df_det)
savefig(plotsdir("sir_det_dynamics.png"))
p_stoch = plot_sir(df_stoch)
savefig(plotsdir("sir_stoch_dynamics.png"))
println("Базовый прогон завершён. Результаты в data/ и plots/")
```

```
using DrWatson
@quickactivate "project"
include(sourcedir("SIRPetri.jl"))
using .SIRPetri
using DataFrames, CSV, Plots
```

Диапазон β

```
β_range = 0.1:0.05:0.8
β_fixed = 0.1
tmax = 100.0
results = []
for β in β_range
```

```

net, u0, _ = build_sir_network( $\beta$ ,  $\beta_{\text{fixed}}$ )
df = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [ $\beta$ ,  $\beta_{\text{fixed}}$ ])
peak_I = maximum(df.I)
final_R = df.R[end]
push!(results, ( $\beta$  =  $\beta$ , peak_I = peak_I, final_R = final_R))
end
df_scan = DataFrame(results)
CSV.write(datadir("sir_scan.csv"), df_scan)

```

График

```

p = plot(
df_scan. $\beta$ ,
[df_scan.peak_I df_scan.final_R],
label = ["Peak I" "Final R"],
marker = :circle,
xlabel = " $\beta$  (infection rate)",
ylabel = "Population",
)
savefig(plotsdir("sir_scan.png"))
println("Сканирование  $\beta$  завершено. Результат в data/sir_scan.csv")

```

5 Выводы

Детерминированная симуляция даёт гладкую, усреднённую динамику эпидемии с классическим пиком заболеваемости.

Стохастическая симуляция (алгоритм Гиллеспи) вносит случайные флуктуации; при большой популяции её результаты близки к детерминированным, но при малых числах различия существенны.

Сканирование параметра β показывает пороговое поведение: при малых β эпидемии практически нет, с ростом β пик инфицированных и конечное число выздоровевших резко возрастают, достигая насыщения.

Анимация динамики наглядно демонстрирует прохождение волны инфекции.

Модель SIR на сетях Петри успешно реализована, позволяет исследовать как среднюю динамику, так и случайные отклонения, а также оценивать чувствительность к параметрам.

Список литературы

Имитационное моделирование. Практикум [https://esystem.rudn.ru/pluginfile.php/3094278/
lab.pdf](https://esystem.rudn.ru/pluginfile.php/3094278/lab.pdf)